

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	14 February 2026
Team ID	LTVIP2026TMIDS54574
Project Name	Smart Sorting Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	4 Marks

Technical Architecture:

TABLE 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface where users upload fruit/vegetable images and view prediction results	HTML, CSS, Flask Templates
2	Application Logic-1	Handles image upload, preprocessing, and sending image to model	Python, Flask

3	Application Logic-2	Performs image preprocessing (resize, normalize, convert to tensor)	Python, Torchvision, PIL
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S.No	Characteristics	Description	Technology
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1	Open-Source Frameworks	Frameworks used to build web application and deep learning model	Flask, PyTorch, Torchvision	
4		Application Logic-3	Performs prediction using trained deep learning model	PyTorch, ResNet18
5		Database	Not required (prediction-based system, no persistent structured data storage)	—
6		Cloud Database	Not used (optional future enhancement)	—
7		File Storage	Stores uploaded images temporarily for prediction	Local File System
8		External API-1	Not used in current implementation	—
9		External API-2	Not used in current implementation	—
10		Machine Learning Model	Classifies fruits and vegetables as Fresh or Rotten using transfer learning	PyTorch, ResNet18
11		Infrastructure (Server / Cloud)	Application runs locally and can be deployed on cloud servers	Local System, Flask Server

TABLE 2: Application Characteristics

2	Security Implementations	Secure handling of uploaded files and controlled local access	Flask File Handling, Local System Security
3	Scalable Architecture	Modular architecture allows easy scaling by deploying on cloud platforms	Flask, Cloud Deployment (AWS / GCP optional)
4	Availability	Application available whenever server is running locally or on cloud	Local Server / Cloud Server
5	Performance	Efficient prediction using optimized deep learning model and GPU support	PyTorch, GPU (optional), Python